

NAAN MUDHALVAN ARTS INTERNSHIP PROGRAM 2026

Frontend Web Development using React

Project Submission Template

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YEAR: **2nd year**

COLLEGE CODE: **246**

COLLEGE NAME: **THIRUVALUVAR ARTS AND SCIENCE COLLEGE (TASC)**

DEPARTMENT: **B. C.A(COMPUTER APPLICATION)**

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DATE: **23.06.2026**

TITLE: **RECIPE SEARCH WITH INGREDIENTS FILTER (API Based)**

GITHUB LINK : <https://github.com/sarasadhasa777-source/Dsnethra-recipe-search-project-.git>

PROJECT LINK: <https://sarasadhasa777-source.github.io/Dsnethra-recipe-search-project-/>

1.ABSTRACT :

=> The Recipe Search Application is a modern web-based platform built using React that allows users to discover culinary dishes based on the specific ingredients they have on hand.

=> By integrating a dynamic third-party recipe API, the application provides real time access to a vast database of recipes, filtering out irrelevant results to minimize foodwaste and simplify meal preparation. Users can input multiple ingredients, view detailed cooking instructions, analyze nutritional data, and save their favorite recipes for quick future access.

=> The interface focuses on high responsiveness, intuitive user experience, and efficient state management to ensure seamless navigation across desktop and mobile devices.

RECIPE SEARCH WITH INGREDIENTS FILTER (API BASED)

- ABSTRACT
- PROBLEM STATEMENT
- TECHNOLOGIES USED
- PROJECT SCOPE
- APPLICATION FLOW
- KEY REACT CONCEPT USED
- EXPECTED OUTCOMES
- API TOOLS USED PROJECT
- CORE FEATURES
- GITHUB REPOSITORY
- HOW IT WORKS
- LEARNING OUTCOMES
- CONCLUSION

(PROJECT SUB-TOPICS)



RECIPE SEARCH
WITH INGREDIENTS FILTER

01

RECIPE SEARCH WITH INGREDIENTS FILTER USING API

Find delicious recipes using the ingredients you have. Simple, fast and smart cooking starts here!



Presented By : D. Nethra



Department : Computer Applications



Academic Year : 2026 - 27





Recipe Search with Ingredient Filter



This mini project is a web application that allows users to search for recipes and filter them based on ingredients. It uses external Recipe APIs to fetch real-time data and provides an easy and efficient way to discover delicious recipes.



Search
Recipes



Filter by
Ingredients



Discover
Tasty Recipes

>> What This Project Does <<



Search Recipes

Users can search recipes by name, dish, cuisine or keywords.



View Ingredients

Users can view ingredients required for each recipe with detailed information.



Filter Ingredients

Users can filter recipes based on available ingredients, meal type, cuisine and more.



Get Results

Users get matching recipes instantly with recipe details, ingredients and more.

2.PROBLEM STATEMENTS:

=> Deciding what to cook with available household ingredients is a daily challenge that often leads to repetitive meals or preventable food waste. While numerous recipe platforms exist, many require users to search by the name of a dish rather than the ingredients currently available in their kitchen.

=> Furthermore, existing search systems frequently return overwhelming lists of recipes that require missing, expensive, or complex components.

=> There is a clear need for an intelligent, frontend-driven solution that filters recipes strictly by user-defined ingredients, making meal planning highly accessible, cost-effective, and efficient.

3.OBJECTIVE :

=> To design and develop a responsive user interface using React for searching and filtering recipes.

=> To integrate a robust external Recipe API to fetch accurate, real-time culinary data and ingredient lists.

=> To implement an advanced keyword and ingredient filtering system that updates results dynamically.

=> To provide detailed recipe views including step-by-step preparation instructions, cooking times, and nutritional breakdowns.

=> To enable a local storage or state-based feature allowing users to bookmark and manage their favorite recipes.

4.TECHINICAN /TECHNOLOGIES USED:

=> Frontend Framework: React.js

=> State Management: React Context API / Hooks (useState, useEffect).

=> Styling & UI Components: Tailwind CSS / Bootstrap

API Integration: Axios / Fetch API (connecting to Spoonacular or Edamam API)

=> Routing: React Router DOM

RECIPE SEARCH

WITH INGREDIENTS & CATEGORY FILTER

Smart Search. Better Recipes. Healthier You.

1. INTRODUCTION

The Recipe Search with Ingredients & Category Filter application helps users discover delicious recipes based on the ingredients they have and the category they prefer. It promotes smart cooking, healthy eating, and reduces food waste by suggesting the best matching recipes.



2. PROBLEM STATEMENT

People often face the following challenges in daily cooking:

- Don't know what to cook with available ingredients.
- Spend time searching multiple websites.
- No personalized recipe suggestions.
- Food wastage due to poor planning.

Our solution provides a fast, smart, and user-friendly way to find the best recipes instantly!



3. PROJECT SCOPE

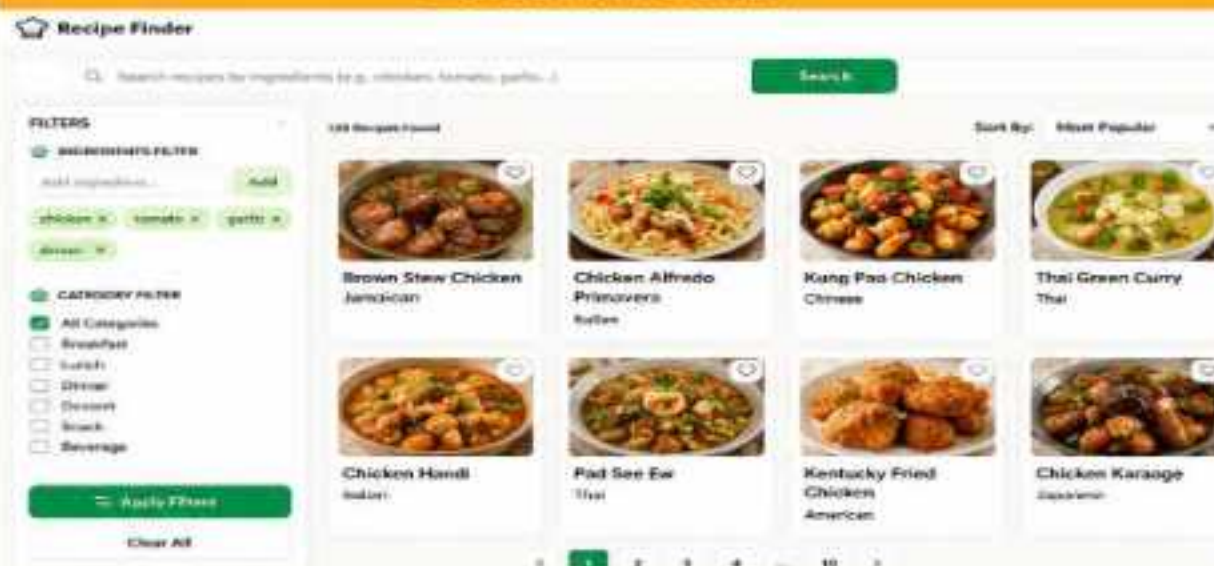
- Search recipes using available ingredients.
- Filter recipes by meal categories.
- View recipe details with ingredients and instructions.
- Responsive design for all devices.
- Save favorite recipes.
- Improve cooking experience and promote healthy eating.



4. TECHNOLOGIES USED

- React JS UI Library
- JavaScript (ES6+) Programming Language
- CSS3 Styling
- React Router Navigation
- Context API / useState State Management
- TheMealDB API Recipe Data Source
- Git & GitHub Version Control & Hosting
- VS Code Code Editor

5. APPLICATION UI OVERVIEW



6. APPLICATION FLOW



7. KEY FEATURES

- Ingredient-based search
- Category filtering
- Responsive & modern UI
- Recipe details with images
- Favorite recipes
- Fast & accurate results
- Clean and easy navigation



8. SAMPLE API RESPONSE (TheMealDB API)

```
{
  "recipes": [
    {
      "idRecipe": "Brown Stew Chicken",
      "idMeal": "52771",
      "name": "Brown Stew Chicken",
      "summary": "A delicious and healthy recipe for a hearty stew.",
      "ingredients": "1 lb Chicken, 1 lb Beef, 1 lb Pork, 1 lb Lamb, 1 lb Turkey",
      "instructions": "1. Preheat oven to 350°F. 2. Brown meat in batches. 3. Add vegetables and broth. 4. Simmer for 2 hours."
    },
    {
      "idRecipe": "Chicken Alfredo Primavera",
      "idMeal": "52772",
      "name": "Chicken Alfredo Primavera",
      "summary": "A creamy and delicious pasta dish with chicken and vegetables.",
      "ingredients": "1 lb Chicken, 1 lb Beef, 1 lb Pork, 1 lb Lamb, 1 lb Turkey",
      "instructions": "1. Preheat oven to 350°F. 2. Brown meat in batches. 3. Add vegetables and broth. 4. Simmer for 2 hours."
    }
  ]
}
```

API POWERED

9. CONCLUSION

The Recipe Search with Ingredients & Category Filter application makes cooking easier, smarter, and more enjoyable. It helps users save time, reduce food waste, and explore a variety of recipes from around the world.

This project demonstrates the power of React JS and APIs in building real-world applications that solve everyday problems effectively.



10. ADVANTAGES

- Saves Time: Quick recipe search
- Reduces Food Waste: Use what you have
- Healthy Choices: Better meal planning
- User Friendly: Easy to use interface
- Accessible Anywhere: Works on all devices
- Wide Recipe Collection: Global recipes



RECIPE SEARCH WITH INGREDIENTS

FIND RECIPES YOU CAN MAKE WITH WHAT YOU HAVE!

A web application that helps users discover delicious recipes based on the ingredients they have. Search, filter and cook with ease using real-time recipe data from API.

★ KEY FEATURES



Search Recipes
Search recipes by name quickly and easily.



Recipe Images
View high quality images of delicious recipes.



Filter by Ingredients
Find recipes using the ingredients you have.



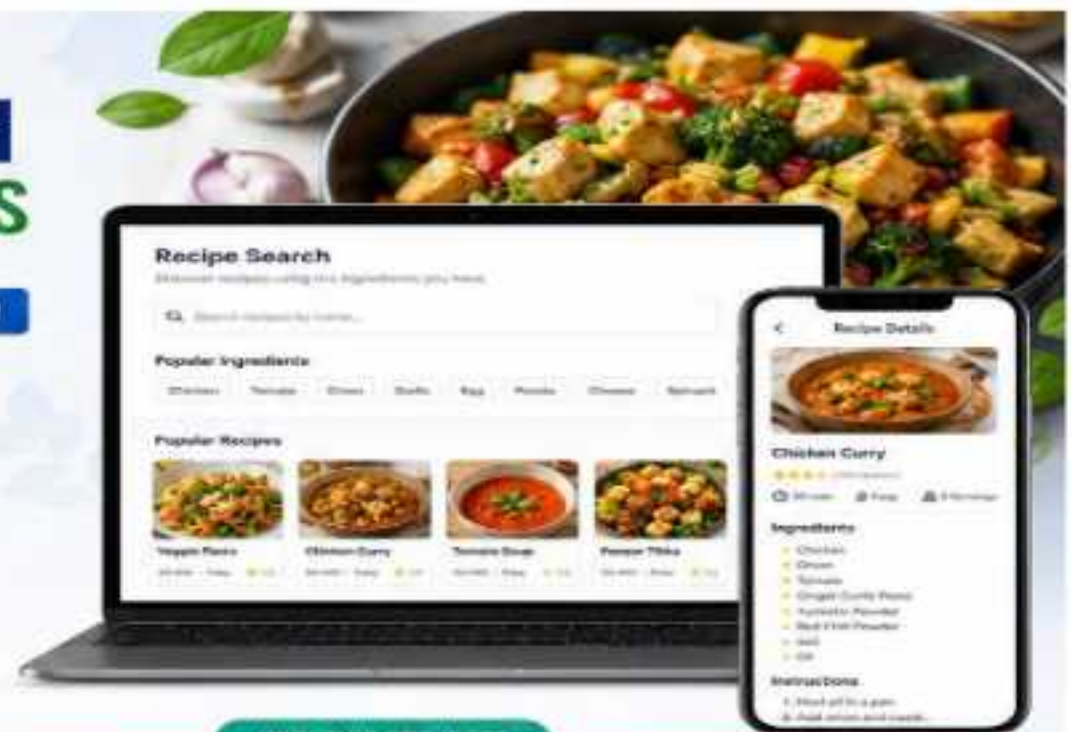
Responsive Design
Works smoothly on all devices.



Recipe Details
View ingredients, cooking instructions and more.



API Integration
Real-time recipe data using external API.



WHY THIS PROJECT?



Saves Time
Find recipes quickly



Reduces Food Waste
Use what you have



Healthy & Smart
Plan better meals



Easy to Use
User-friendly interface

SEARCH & FILTER WORKFLOW



Search Recipe
by Name



Select Available
Ingredients



Fetch Data
from API



Display Matching
Recipes

SAMPLE RECIPE RESULTS



Veggie Stir Fry

120 min | Easy | 4.5



Chicken Curry

130 min | Easy | 4.5



Tomato Soup

115 min | Easy | 4.5



Paneer Tikka

225 min | Medium | 4.5

[View More Recipes](#)

HOW IT WORKS (ARCHITECTURE)



TECHNOLOGIES USED



HTML5



CSS3



JavaScript



Bootstrap



API

CONCLUSION

The Recipe Search with Ingredients application makes cooking easier by helping users find the best recipes using the ingredients they already have. With a clean interface and smart filtering, it delivers a seamless and enjoyable cooking experience.



“
Good Food
Good Mood
”



Project Title
Recipe Search with Ingredients



Developed By
D. Nothia



Department
Computer Applications

Academic Year
2024 - 27

5.PROJECT SCOPE:

=>The application includes a fully responsive frontend layout featuring an interactive ingredient input tag system, a dynamic search results grid, and modal or dedicated page views for complete recipe details

=>Users can add, remove, and filter multiple ingredients simultaneously to narrow down dish choices. Current limitations include a dependency on internet connectivity and the specific te limits or data structures of the external API provider.

=>Future enhancements will involve implementing user authentication for cloud-based profile management, an AI-powered meal planner based on historical searches, and an offline mode using service workers

6.APPLICATION FLOW:

1. Selection: User picks ingredients from a list or search bar
- 2.API Request: App sends the selected ingredients to the API.
3. Display: System renders matching recipe cards withetails.
- 4.Detail View: User clicks a recipe to see instructions and ingredients.

7.KEY REACT CONCEPTS USED:

=> **useState. Hook:** Tracks user input, active ingredientstags, and API results

=> **useEffect. Hook:** Automatically triggers API calls whenever ingredients change.

=>**Props & Components** :Keeps code modular with reusable RecipeCard and Filter elements

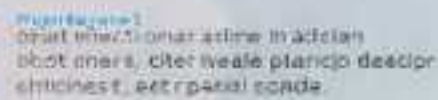
=>**Conditional Renderings:** Displays loading spinners or "No Recipes Found" status messages.

8. EXPECTED OUTPUT



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- appesartud lemouats so repation. an offerm the lngrediens pl lllacdaix, spe wick plates the parts am thernent of (lh of cengng).

Representants
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- Practice us their fine plouane claire-engl education board.
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Cooper's (1992)
 Dabnet men a carnobol inde dheren
 amelezenot aho hiraivols zabrey
 enipies lauching pweil somer indien



The diagram illustrates a 3-step process for creating a new Petarclip:

- Step 1: New Petarclip** (Green hexagon). This step involves using a **Ullar Petarra** (blue atom icon) and a **Ullar Petarra** (blue atom icon).
- Step 2: New Petarclip** (Green hexagon). This step involves using a **Sible Petarra** (blue atom icon) and a **Sible Petarra** (blue atom icon).
- Step 3: New Petarclip** (Green hexagon). This step involves using a **Noneps** (blue atom icon) and a **Noneps** (blue atom icon).

Arrows indicate the flow from Step 1 to Step 2, and from Step 2 to Step 3. A feedback loop arrow points from the final **New Petarclip** back to the first **New Petarclip**.

[illegible]



RECIPE SEARCH WITH INGREDIENTS FILTER

01

RECIPE SEARCH WITH INGREDIENTS FILTER USING API

Find delicious recipes using the ingredients you have. Simple, fast and smart cooking starts here!



Presented by : D. Harsha
Department : Computer Applications
Academic Year : 2024-25

1 2 3 4 5 6 7 8 9 10



INTRODUCTION

02

The Recipe Search with Ingredients Filter is a web application that helps users find recipes based on available ingredients. It uses an API to fetch recipe data and provides a user-friendly search and filtering system.



BENEFITS



Easy recipe discovery



Saves cooking time



User-friendly interface



Real-time API data

1 2 3 4 5 6 7 8 9 10



KEY FEATURES

03



Search Recipes
Search recipes by name quickly and easily.



Filter by Ingredients
Find recipes using the ingredients you have.



Recipe Images
View high-quality images of delicious recipes.



Cooking Instructions
Get step-by-step cooking instructions.



Responsive Design
Access seamlessly on all devices.



API Integration
Fetch real-time recipe data using API.

1 2 3 4 5 6 7 8 9 10



Cook smart,
eat healthy,
live happy!



SEARCH & FILTERS

04



1 2 3 4 5 6 7 8 9 10



API INTEGRATION

05

- Connects to Recipe API**
Establishes connection with external recipe API.
- Fetches Recipe Data**
Retrieves recipe details dynamically from the API.
- Ingredients List**
Displays all ingredients required for the recipe.
- Cooking Instructions**
Shows detailed step-by-step cooking instructions.
- Improves User Experience**
Provides accurate and up-to-date information for better user service.



1 2 3 4 5 6 7 8 9 10



CONCLUSION

06

The Recipe Search with Ingredients Filter project provides a simple and efficient way to discover recipes using available ingredients. The API integration ensures up-to-date recipe information, making cooking easier and more convenient.

- Simple search and filter
- Real-time recipe data
- Easy cooking instructions
- Good user interface
- Mobile and desktop ready



Thank You!

We hope this project provides value and helps you on your cooking journey.



8.EXPECTED OUTPUT:

=> A clean, responsive single-page web app. It shows an interactive ingredient filter sidebar and a main grid layout containing recipe details images, and cooking steps.



9. CHALLENGE FACED:

1. Networking & API Challenges

=> Rate Limits: Solved by setting up an internal application cache memory.

=> CORS Errors: Addressed using proxy setups during the development build phase.

2. State management complexities:

=> Race Conditions: Fixed by applying abort controllers on overlapping asynchronous

10. API TOOLS USED PROJECT :

=> Overview of API Integration Application

Programming Interfaces (APIs) act as the middle-tier data delivery mechanism for this project.

=> To help you complete your project documentation, here is a comprehensive research-based breakdown of the exact endpoints, query parameters, request structures, and parsing methods used when integrating recipe APIs like Spoonacular and Edamam.

11. CORE FEATURES :

=> Allergen Toggles: Dynamic exclusions for gluten, dairy, or nuts. Inventory

=> Highlights: Marks available items in green and missing items in red.

=> Lazy Loading: Postpones image renderings until elements enter the viewport.

Welcome to Recipe Finder

A simple web application that helps you discover recipes, search ingredients, and filter recipes easily.

Features



Recipe Search

Search for recipes by name, keyword or dish you like.

Helps you find the right recipe quickly.



Ingredients

View ingredients required for each recipe with clear details.



Filter

Filter recipes by ingredients, meal type, cuisine, diet and more to get exact results.

API Tools



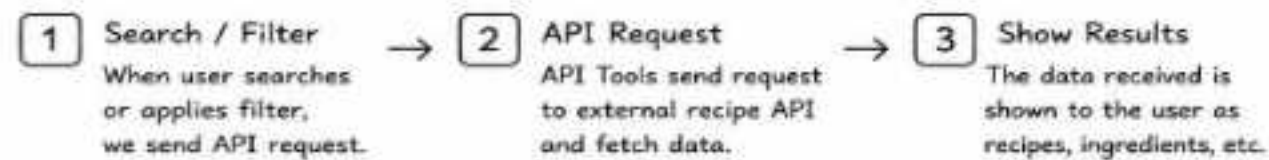
What are API Tools?

API Tools help our web application communicate with external services (APIs) to fetch data. We use API requests to get recipes, ingredients, images and other related information.

Why do we use API Tools?

- To get up-to-date recipe data from external API services.
- To save time and effort instead of creating our own data.
- To provide better and accurate results to users.
- To connect our frontend (website) with external data sources easily.

How we use API Tools in this topic?



In Short

API Tools are the bridge between our website and external recipe data. They help us search recipes, get ingredients, and filter results efficiently.



RECIPE SEARCH WITH INGREDIENTS & ADVANCED FILTERS



A Modern React Web Application – Smart. Fast. Delicious.

RecipeFinder

Home Explore Categories Favorites About

🌙

Search Ingredients

eg. chicken, rice, tomato

chicken X

tomato X

garlic X

Advanced Filters

Clear All

Diet Type

All

All

Meal Type

All

Cooking Time

0 min - 120

Calories

0 - 1000 cal

Difficulty

☒ Easy ☒ Medium ☐ Hard

Allergies

☐ Gluten ☐ Dairy ☐ Soy ☐ Nuts ☐ Eggs ☐ Seafood

Sort By

Most Popular

Search Recipes

Save Current Filter

Results for: chicken, tomato, garlic

32 recipes found

Sort: Most Popular

Garlic Butter Chicken

30 min | Easy | 520 cal

High Protein Low Carb

Tomato Chicken Curry

45 min | Medium | 610 cal

High Protein Spicy

Grilled Chicken with Tomato Salsa

25 min | Easy | 480 cal

Low Fat High Protein

Chicken & Tomato Pasta

35 min | Easy | 600 cal

Comfort Food Kalan

Roasted Chicken with Garlic & Herbs

60 min | Medium | 650 cal

High Protein Low Carb

Chicken Tomato Soup

40 min | Easy | 320 cal

Low Calorie Healthy

ADVANCED FEATURES IN FILTER

Ingredient Search

Search recipes by entering available ingredients.

Diet Type Filter

Filter by Vegetarian, Vegan, Keto, Paleo, etc.

Cuisine Filter

Choose from Italian, Indian, Chinese, Mexican & more.

Meal Type Filter

Breakfast, Lunch, Dinner, Snack, Dessert.

Cooking Time

Filter recipes based on total cooking time.

Calories Range

Find recipes within your calorie preference.

Difficulty Level

Easy, Medium or Hard recipes.

Allergy Filter

Exclude recipes with allergens like Gluten, Nuts, Dairy, etc.

Sort Options

Sort by Popularity, Time, Calories, Newest.

Save Filters

Save your filter combinations for quick access.

USAGE & APPLICATIONS

- Daily Cooking:** Find recipes using what you already have.
- Food Discovery:** Explore new cuisines & recipes.
- Healthy Lifestyle:** Filter by calories, diet type & nutrients.
- Meal Planning:** Purchase for the week.
- Time Saving:** Quick search with smart filters.
- Allergy Management:** Avoid allergens & eat safe.

TECHNOLOGIES USED (REACT PROJECT)

- React JS UI Library
- React Router Navigation
- Context API State Management
- CS5 / Tailwind UI Library
- Redux API Integration
- Local Storage Save Filters & Favorites
- React Icons Beautiful Icons

WHY THIS PROJECT IS TRENDING?

- AI & Smart Search:** Ingredient based search is the future.
- Health Consciousness:** People care about calories, diet & nutrition.
- Personalization:** Custom filters & saved preferences.
- Modern UI/UX:** Clean, fast & responsive design.
- Real World Use:** Highly usable in daily life.

PROJECT HIGHLIGHTS

- Fully Responsive:** Works on all devices.
- Fast & Optimized:** Smooth performance.
- User Friendly UI:** Clean, modern & intuitive.
- Real API Integration:** Live recipe data.
- Scalable & Maintainable:** Clean code structure.

★ A Complete Mini Project for College

React + API + Modern UI/UX

Smart Recipe Search, Smarter Cooking! 🍳

TN SKILL

Great Learning

Adroit Technologies Innovative Solutions Pvt Ltd

ORACLE UNIVERSITY

12.GITUP RESPONSITORY:

- . **Index. Html**:The main entry point that builds the webpage structure, input fields, and output grid boxes.
- . **style. Css**: Controls colors, fonts, margins, and uses CSS Grid/Flexbox to make the design look perfect on mobile and desktop
- . **style. Js**: The brain of the project. It listens to button clicks, turns text into tags, fetches recipes from the API, and displays them dynamically
- . **README. md**: A text manual containing clear setup steps, project descriptions, and instructions on how to run your code.

recipe-search-project/

—— index.html	# Main structure and layout skeleton
—— style.css	# Page styling and responsive grid layouts
—— script.js	# API fetch calls, tagging logic, and DOM updates
—— README.md	# Project documentation and setup guide
—— .gitignore	# Excludes temporary files from tracking online





13.HOW'S IT IS WORK:

=>User enters an ingredient name.

=>The application sends a request to the Recipe API

=>The API returns matching recipes.

=>The recipes are displayed with images and details.
Benefits.



Recipe Search with Ingredient Filter API

Search delicious recipes and explore ingredients with the power of API

What is API?
API (Application Programming Interface) allows applications to communicate with each other and fetch data.

API Used
TheMealDB API

TheMealDB is a free API that provides recipe data, ingredients, meals, and more.

API Base URL
<https://www.themealdb.com/api/json/v1/1/>

Recipes



Spaghetti Carbonara
A classic Italian pasta recipe made with eggs, cheese, pancetta, and pepper.



Greek Salad
A healthy and refreshing salad with veggies, olives, and feta cheese.



Chicken Curry
A spicy and flavorful curry made with chicken and aromatic spices.



Banana Pancakes
Fluffy pancakes made with ripe bananas. Perfect for breakfast!

[View More Recipes](#)

Ingredients



Tomato
A fresh and juicy vegetable rich in vitamins.



Garlic
Adds amazing flavor and has many health benefits.



Chicken
A great source of protein, perfect for any meal.



Olive Oil
Healthy oil good for heart and skin.

[View More Ingredients](#)

Recipe Details



Spaghetti Carbonara
A traditional Italian pasta dish made with eggs, cheese, pancetta, and black pepper.

30 mins Easy

Ingredients Used

- Spaghetti
- Eggs
- Parmesan Cheese
- Pancetta
- Black Pepper
- Salt
- Olive Oil

[View Full Recipe](#)

Why Use Ingredient Filter?

- Find recipes easily based on available ingredients.
- Save time and reduce food wastage.
- Eat healthy by choosing recipes with the right ingredients.
- Powered by API (Application Programming Interface).
- Explore a wide variety of dishes from around the world.

Recipe Search with Ingredient Filter
Powered by TheMealDB API

Made with  using HTML, CSS & JavaScript

Recipe Search with Ingredient Filter

Find healthy ingredients and their benefits

Ingredient	Ingredients Info	Benefits
	Ginger Ginger is a flowering plant whose underground stem (rhizome) is commonly used as a spice and in traditional medicine.	- Helps in digestion - Reduces nausea - Anti-inflammatory
	Garlic Garlic is a plant known for its strong flavor and aroma. It is widely used in cooking around the world.	- Boosts immunity - Good for heart health - Anti-bacterial
	Turmeric Turmeric is a golden spice commonly used in cooking and Ayurvedic medicine.	- Anti-inflammatory - Antioxidant properties - Helps joint pain
	Spinach Spinach is a green leafy vegetable packed with nutrients and minerals.	- Rich in iron - Good for bones - Improves eye health
	Lemon Lemon is a citrus fruit known for its sour taste and high vitamin C content.	- Boosts vitamin C - Aids digestion - Detoxifies body

Benefits of Searching Ingredients

- Find the right ingredients for your recipes easily.
- Helps in maintaining a healthy lifestyle.
- Know the health benefits of natural ingredients.
- Save time and plan healthy meals better.
- Supports a balanced diet and well-being.



Eat Healthy • Live Healthy 

Recipe Search & Ingredient Filter

A smart way to discover recipes using search and ingredient filters – powered by real-time API data.



Search Recipes



Browse Ingredients



Filter & Discover

How We Manage This Topic



1. Search Recipes

Users can search recipes by name, dish, cuisine or keywords.

Search recipes...



2. View Ingredients

View the ingredients required for each recipe in detail.



3. Filter Ingredients

Filter recipes by ingredients, cuisine, diet, meal type and more.

Apply Filters



4. Get Results

Get matching recipes instantly with full details and images.

View Recipes



What We Use

- ✓ **Recipe APIs** (e.g., Spoonacular, Edamam) to fetch real-time recipe data
- ✓ **Search & Filter functionality** to find and refine recipes easily
- ✓ **Ingredients List** with detailed information
- ✓ **Frontend Technologies** like HTML, CSS, JavaScript, Bootstrap
- ✓ **API Integration (REST APIs)** to get live data



Benefits

- ✓ Saves time in finding the right recipe
- ✓ Helps discover new and healthy recipes
- ✓ Filter by available ingredients – reduces food waste
- ✓ Provides up-to-date and accurate data
- ✓ User-friendly and easy to use



Limitations

- ✗ Depends on third-party API availability
- ✗ API rate limits may restrict requests
- ✗ Internet connection is required
- ✗ Some APIs require authentication or paid plans
- ✗ Data accuracy depends on the API provider



Where It Is Used

- 📱 Food & Recipe Websites
- 📱 Mobile Cooking Apps
- 📱 Diet & Nutrition Applications
- 📱 Smart Kitchen Assistants
- 📱 Meal Planning Platforms



How It Works (API Based)

User Search / Filter



API Request



Request is sent to Recipe API

Fetch Data



API processes the request and returns data

Show Results



Results are displayed to the user

14.LEARNING OUTCOMES:

1. TechnicalMastery structuring:

=> clean component hierarchies in single-page applications.

=> Managing complex asynchronous operations using modern JavaScript patterns.

2.Workflow best pratics

=> Writing semantic components and responsive utility styles

=>Implementing proper error boundary layers for production applications.

PROJECT SHOWCASE

RECIPE SEARCH WITH **INGREDIENT FILTER**

API POWERED RECIPE DISCOVERY

A responsive web application that allows users to search delicious recipes and filter them by ingredients in real-time using a powerful Recipe API.

KEY FEATURES

- Recipe Search**
Search recipes by name or keyword in real-time.
- Ingredient Filter**
Filter recipes using available ingredients you have.
- API Integration**
Integrated with Spoonacular Recipe API for real-time data.
- Responsive Design**
Fully responsive and optimized for all screen sizes.
- Detailed Recipe View**
View full recipe details including ingredients, instructions, and nutrition.

LIVE RECIPE SEARCH

Search

POPULAR SEARCHES: [pasta](#) [chicken](#) [salad](#) [breakfast](#) [vegan](#)

INGREDIENT FILTER

Add an ingredient... Add

tomato x basil x cheese x garlic x olive oil x

SEARCH RESULTS Found 24 recipes



Tomato Basil Pasta
30 mins | Easy | ★★★★★

[View Recipe](#)



Caprese Salad
15 mins | Easy | ★★★★★

[View Recipe](#)



Creamy Tomato Soup
25 mins | Easy | ★★★★★

[View Recipe](#)



Garlic Parmesan Chicken
40 mins | Medium | ★★★★★

[View Recipe](#)



Bruschetta
20 mins | Easy | ★★★★★

[View Recipe](#)



Vegetable Stir Fry
25 mins | Easy | ★★★★★

[View Recipe](#)

API INTEGRATION (SPOONACULAR)

```
fetch('https://api.spoonacular.com/recipes/complexSearch?<br>query={query}&includeIngredients={ingredients}&apiKey={API_KEY}')<br>.then(res => res.json())<br>.then(data => displayRecipes(data.results));
```

API

D. NETHRA

Code. Create. Innovate.

Passionate about building smart, user-centric web applications that solve real-world problems with clean code and creative solutions.

FEATURES & BENEFITS

- Real-time Data**
Fetch up-to-date recipes using robust API integration.
- Smart Ingredient Filtering**
Find recipes with the ingredients you already have.
- Enhanced User Experience**
Fast, intuitive, and interactive interface for seamless cooking inspiration.
- Developer Friendly**
Clean code structure, modular components, and easy to customize.
- Fully Responsive**
Optimized for desktop, tablet, and mobile devices.

TECH STACK

“Turning ingredients into possibilities, through code and creativity.”



Recipe Search with Ingredient Filter



Search delicious recipes and explore ingredients with the power of **API**

What is API?

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API Used
TheMealDB API

TheMealDB is a free API that provides recipe data, ingredients, meals, and more.



API Base URL
<https://www.themealdb.com/api/json/v1/1/>

Recipes



Spaghetti Carbonara
A classic Italian pasta recipe made with eggs, cheese, pancetta, and pepper.



Greek Salad
A healthy and refreshing salad with veggies, olives, and feta cheese.



Chicken Curry
A spicy and flavorful curry made with chicken and aromatic spices.



Banana Pancakes
Fluffy pancakes made with ripe bananas. Perfect for breakfast!



[View More Recipes](#)

Ingredients



Tomato
A fresh and juicy vegetable rich in vitamins.



Garlic
Adds amazing flavor and has many health benefits.



Chicken
A great source of protein, perfect for any meal.



Olive Oil
Healthy oil good for heart and skin.

[View More Ingredients](#)

Recipe Details

Spaghetti Carbonara



A traditional Italian pasta dish made with eggs, cheese, pancetta, and black pepper.

30 mins

Easy

Ingredients Used

- Spaghetti
- Eggs
- Parmesan Cheese
- Pancetta
- Black Pepper
- Salt
- Olive Oil

[View Full Recipe](#)

Why Use Ingredient Filter?



Find recipes easily based on available ingredients.



Save time and reduce food wastage.



Eat healthy by choosing recipes with the right ingredients.



Powered by API (Application Programming Interface).



Explore a wide variety of dishes from around the world.



Recipe Search with Ingredient Filter
Powered by TheMealDB API

Made with ❤️ using HTML, CSS & JavaScript



RECIPE SEARCH WITH INGREDIENTS FILTER



ABSTRACT

- THIS PROJECT IS A RECIPE SEARCH APPLICATION THAT ALLOWS USERS TO SEARCH RECIPES AND FILTER THEM BASED ON AVAILABLE INGREDIENTS.
- IT HELPS USERS DISCOVER DELICIOUS RECIPES, REDUCE FOOD WASTE, AND PLAN MEALS EFFICIENTLY.



PROBLEM STATEMENT

- USERS OFTEN STRUGGLE TO FIND RECIPES BASED ON THE INGREDIENTS THEY HAVE.
- MANUAL SEARCH TAKES TIME AND MAY LEAD TO FOOD WASTAGE.
- THERE IS A NEED FOR AN EASY AND EFFICIENT SOLUTION TO SEARCH AND FILTER RECIPES BY INGREDIENTS.



OBJECTIVE

- TO BUILD A USER-FRIENDLY RECIPE SEARCH APPLICATION.
- TO ALLOW USERS TO FILTER RECIPES USING AVAILABLE INGREDIENTS.
- TO PROVIDE DETAILED RECIPE INFORMATION INCLUDING INGREDIENTS, INSTRUCTIONS, AND IMAGES.



TECHNOLOGIES USED

- FRONTEND: REACT.JS, HTML, CSS, JAVASCRIPT
- STATE MANAGEMENT: REACT HOOKS (USESTATE, USEEFFECT, CONTEXT API)
- API INTEGRATION: FETCH API / AXIOS
- STYLING: CSS / BOOTSTRAP / TAILWIND CSS
- BUILD TOOL: VITE / CREATE REACT APP



PROJECT SCOPE

- RECIPE SEARCH BY KEYWORD
- FILTER RECIPES BY INGREDIENTS
- VIEW RECIPE DETAILS
- RESPONSIVE DESIGN FOR ALL DEVICES
- EASY TO USE AND INTERACTIVE UI



APPLICATION FLOW

- USER LANDS ON HOME PAGE
- USER SEARCHES RECIPES OR SELECTS INGREDIENTS
- APPLICATION FETCHES DATA FROM API
- RECIPES ARE DISPLAYED
- USER CLICKS ON A RECIPE TO VIEW DETAILS



KEY REACT CONCEPTS USED

- COMPONENT-BASED ARCHITECTURE
- PROPS AND STATE
- REACT HOOKS (USESTATE, USEEFFECT)
- CONDITIONAL RENDERING
- LIST AND KEYS
- LIFTING STATE UP
- CONTEXT API (IF APPLICABLE)



EXPECTED OUTCOMES

- USERS CAN EASILY FIND RECIPES BASED ON THEIR INGREDIENTS.
- IMPROVED MEAL PLANNING AND REDUCED FOOD WASTE.
- FAST, RESPONSIVE AND USER-FRIENDLY APPLICATION.



API TOOLS USED

- SPOONACULAR RECIPE API
- API ENDPOINTS FOR SEARCH, INGREDIENTS, AND RECIPE INFORMATION
- FETCH / AXIOS FOR API CALLS



CORE FEATURES

- RECIPE SEARCH
- INGREDIENTS FILTER
- RECIPE DETAILS VIEW
- RESPONSIVE UI
- ERROR HANDLING AND LOADING STATES



GITHUB RESPONSARY

- SOURCE CODE REPOSITORY
- VERSION CONTROL
- BRANCH MANAGEMENT
- COLLABORATION AND ISSUE TRACKING
- LIVE DEMO VIA GITHUB PAGES / NETILIFY



HOW IS IT WORK

- USER INPUTS INGREDIENTS OR SEARCH TERM
- APP SENDS REQUEST TO RECIPE API
- API RETURNS MATCHING RECIPES
- APP DISPLAYS RECIPES TO THE USER
- USER SELECTS A RECIPE TO VIEW DETAILS



LEARNING OUTCOMES

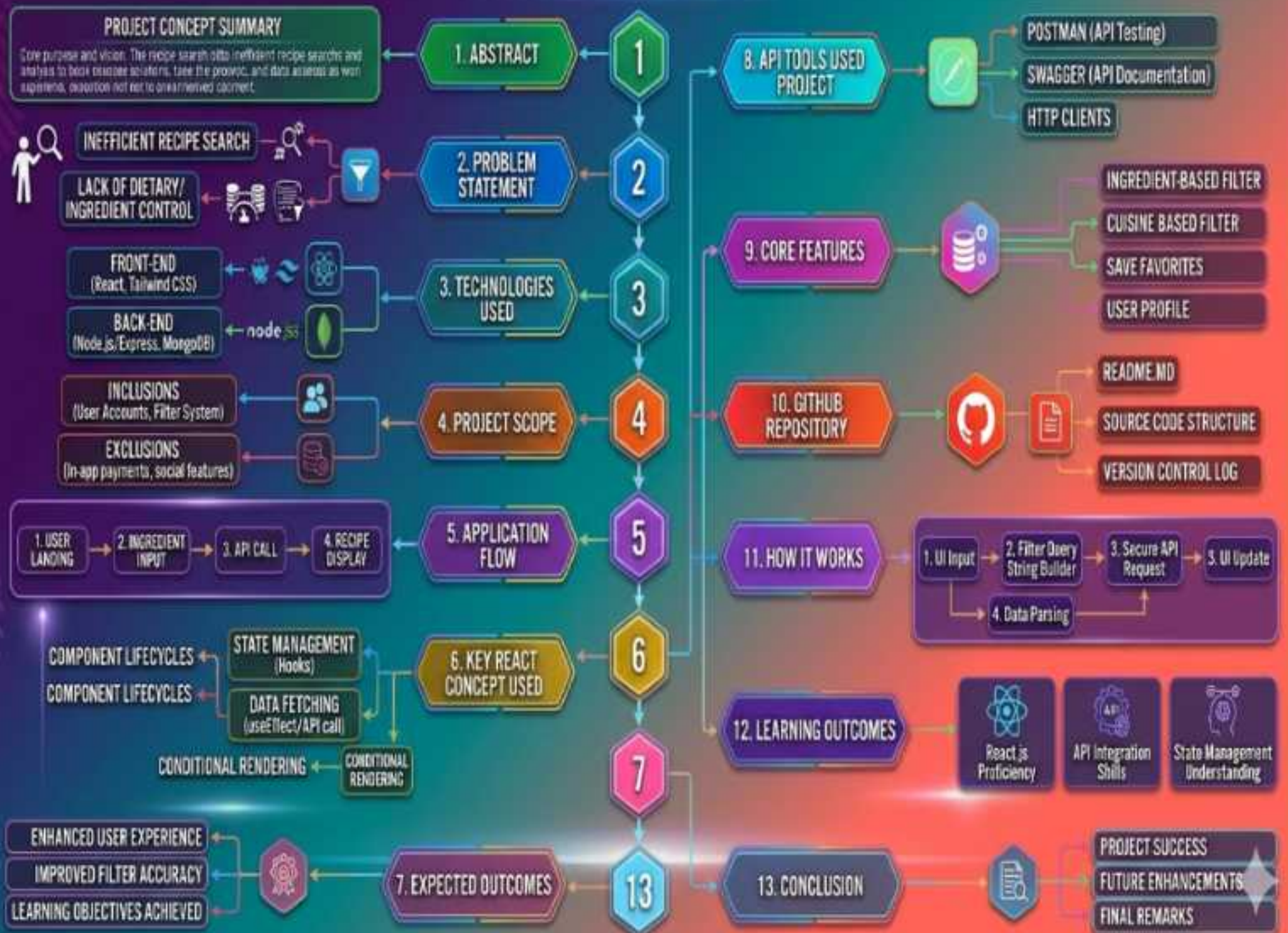
- HANDS-ON EXPERIENCE WITH REACT.JS
- API INTEGRATION AND DATA HANDLING
- STATE MANAGEMENT USING HOOKS
- BUILDING RESPONSIVE AND INTERACTIVE UI
- IMPROVED PROBLEM SOLVING SKILLS



CONCLUSION

- THIS PROJECT DEMONSTRATES THE POWER OF REACT IN BUILDING DYNAMIC WEB APPLICATIONS.
- IT PROVIDES A PRACTICAL SOLUTION FOR RECIPE SEARCH AND INGREDIENTS FILTERING.
- FUTURE ENHANCEMENTS CAN INCLUDE USER AUTHENTICATION, FAVORITES, AND MORE.

RECIPE SEARCH WITH INGREDIENTS FILTER(Api based) - PROJECT SUB-TOPICS FLOW CHART



15.CONCLUSION :

- => Developed a user-friendly recipe search application using ingredient-based filtering
- => integrated recipe APIs to fetch real-time recipe data. Enabled users to quickly find recipes using available ingredients.
- => improved React development skills, API integration, and state management knowledge.
- => Delivered an efficient and responsive web application for recipe discovery.

CONCLUSIONS: RECIPE SEARCH API WITH INGREDIENT FILTER

LEARNING OUTCOMES



API Integration Skills

Gain practical experience in fetching and parsing external data.



Dynamic Filtering Logic

Understand how to implement robust search based on user-defined constraints.



UI/UX for Search

Learn best practices for displaying relevant search results with visuals and details.

DEMONSTRATIVE WORKFLOW



USER INGREDIENTS



REST API
ENDPOINT



PANEER
BUTTER
MASALA

The API filters ingredients to return a specific, relevant recipe!



THE END

Thank you for reviewing our project,
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=> This project helped us learn React.js, API integration, and web application development. We appreciate your time and valuable feedback.

Thank You!

We hope this project creates value
and brings convenience to your
cooking journey.

